TACO: TabPFN Augmented Causal Outcomes for Early Detection of Long COVID Supplementary Materials

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1 Data Preprocessing Pipeline

1.1 RNA-seq Data

From the raw RNA-seq dataset, we:

1. Removed rows and columns containing only NA values.
2. Transposed the table so genes became columns and samples became rows.
3. Extracted `Subject_ID` from sample names (splitting at the first "T").
4. Averaged gene expression values across multiple samples per subject.

This yielded **489** subject-level profiles for downstream modelling. Class distribution after merging labels was **0 = 186 (38.0%)** vs **1 = 303 (62.0%)**.

1.2 Clinical Data Processing

We merged current and historical symptom variables (e.g., `Symptom_X` and `Symptom_X_Ever`) so that a symptom was marked positive if present in either column. Long COVID status was assigned based on any of:

- Presence of at least one persistent post-COVID symptom,
- Recovery status marked as "Not Recovered",
- Post-COVID health reported as "Worse".

We then merged the clinical labels with the RNA-seq dataset using `Subject_ID` as the key.

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1.3 Most Variable Genes (MVG)

For benchmark models, we selected the **500** genes with the highest variance across subjects:

$$\mathrm{Var}(g_i) = \frac{1}{n-1}\sum_{j=1}^{n}(x_{ij} - \bar{x}_i)^2,$$

a standard, non-causal baseline.

2 Differential Causal Effects (DCE) for TACO

We constructed directed pathway graphs from KEGG [3] using ECrel (enzyme–enzyme) and GErel (gene expression) edges, excluding PPrel. Using the 489 profiles, we estimated differential causal effects per source–target edge across disease states, applied FDR correction (q<0.05), and prioritised **411** genes with significant causal influence [2]:

$$G_{\mathrm{TACO}} = \{g_i : |\beta_i| > \tau,\ \mathrm{FDR}_i < 0.05\}.$$

These genes define TACO's causal feature set.

3 Evaluation Protocol and Metrics

3.1 Cross-validation

We used **5-fold stratified CV** (single repetition, `random_state=42`). All models were trained and evaluated identically on:

- **DCE (411)** causal genes, and
- **MVG (500)** most-variable genes.

3.2 Core and Extended Metrics

Let TP, TN, FP, FN be confusion-matrix counts; $\hat{p}$ positive-class probability.

$$\mathrm{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \mathrm{Precision} = \frac{TP}{TP+FP}, \quad \mathrm{Recall} = \frac{TP}{TP+FN}, \tag{1}$$

$$\mathrm{F1} = \frac{2\cdot\mathrm{Precision}\cdot\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}, \quad \mathrm{ROC\ AUC} = \int_0^1 \mathrm{TPR}(t)\,dt. \tag{2}$$

See [1] for ROC analysis and [4] for classification metrics.

Extended metrics used in the code and saved to CSV:

$$\text{Balanced Acc} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right), \quad \text{Specificity} = \frac{TN}{TN+FP}, \quad \text{NPV} = \frac{TN}{TN+FN}, \tag{3}$$

$$\text{PR AUC} = \int_0^1 \text{Precision}(\text{Recall})\, d\text{Recall}, \quad \text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{p}_i)^2, \tag{4}$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}, \quad \kappa = \text{Cohen's kappa}. \tag{5}$$

4 Models and Ensembles

4.1 TACO (TabPFN with DCE context)

TACO uses the **411 DCE** genes as features and a pre-trained Transformer (TabPFN) for training-free Bayesian-style inference on tabular data.

4.2 Benchmarks (MVG-500)

We compared against linear (Logistic Regression, Ridge, LDA), kernel (SVC), neighbors (KNN), neural (MLP), and tree ensembles (Random Forest, Extra-Trees, Gradient Boosting, AdaBoost), all on the **MVG-500** set.

4.3 Post-hoc Ensemble (Top-5 only)

We use a single probability-based ensemble, `Ensemble_TabPFN_Top5`. The ensemble members are the *top five* individual models selected within each run by mean F1 across CV folds. For each test point x_i, we extract a positive-class probability from each model m using a safe fallback:

$$\hat{p}_i^{(m)} = \begin{cases} \texttt{predict\_proba}(x_i)_{[1]}, & \text{if available,} \\ \sigma(\texttt{decision\_function}(x_i)), & \text{else if available,} \\ \mathbf{1}\{\texttt{predict}(x_i) = 1\}, & \text{otherwise,} \end{cases} \quad \text{with } \sigma(z) = \frac{1}{1+e^{-z}}.$$

Soft voting averages these probabilities across the Top-5 models:

$$\hat{p}_i = \frac{1}{K}\sum_{m \in \mathcal{M}_{\text{Top5}}} \hat{p}_i^{(m)}, \qquad \hat{y}_i = \mathbf{1}\{\hat{p}_i \geq 0.5\},$$

where $K = 5$. This produces the final predicted label, which we report in the CV summaries. No additional weighting or external calibration is applied beyond the above fallback to probabilities.

5 Implementation and Reproducibility

5.1 Environment

- Python 3.10/3.11 (as available)
- **TabPFN 2.2.1**; PyTorch device (effective): `cpu`
- scikit-learn, numpy, pandas, **statsmodels** (for significance testing)

5.2 Run Artifacts

The notebook saves per-fold results, full metric summaries (including extended metrics), and manifests to run-specific output folders. Tables shown above are the exact means $\pm$ SD printed by the notebook.

The complete TACO framework implementation is available at: https://github.com/SindyPin/TACO

References

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